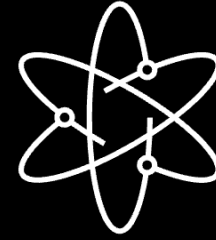


Practical Low-Code Programming with Node-RED

Kazuhito Yokoi
Software Engineer,
Node-RED project

Orlando, FL
October 6-9



Explore.
Build.
Launch.

GO/

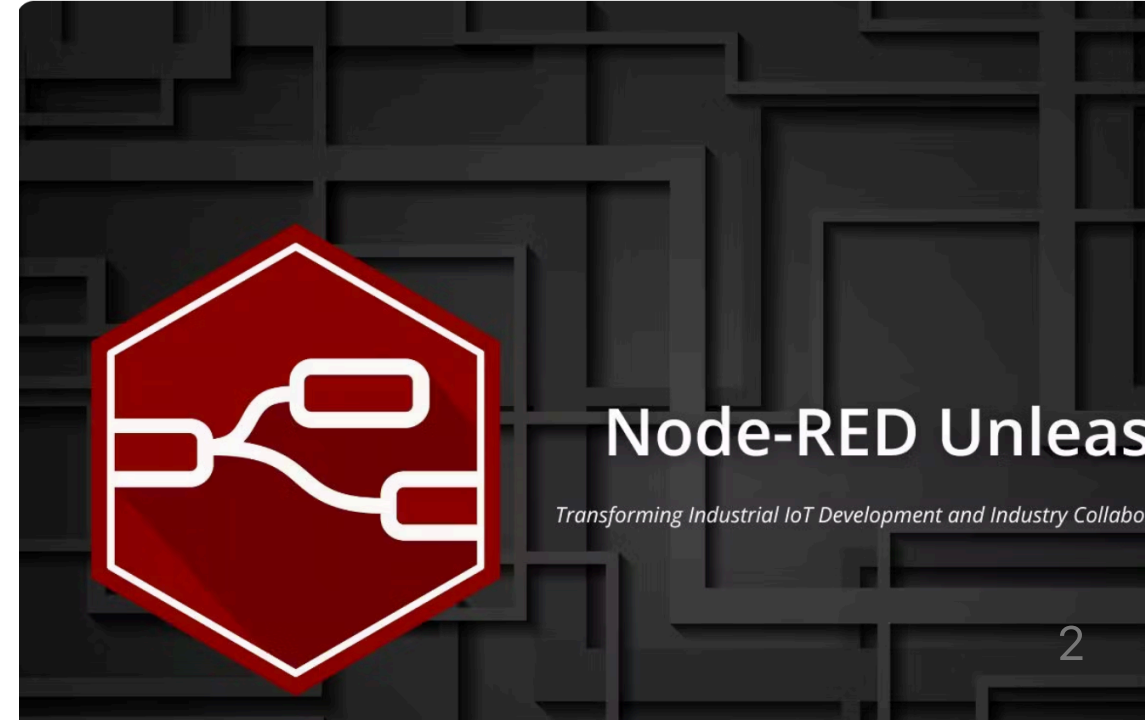


Projects

Node-RED Unleashed: Transforming Industrial IoT Development and Industry Collaboration with Hitachi

Posted On: March 27, 2024

Hear from Hitachi's Kazuhito Yokoi to find out more about Node-RED and why it is becoming so popular in Industrial IoT applications.



Kazuhito Yokoi

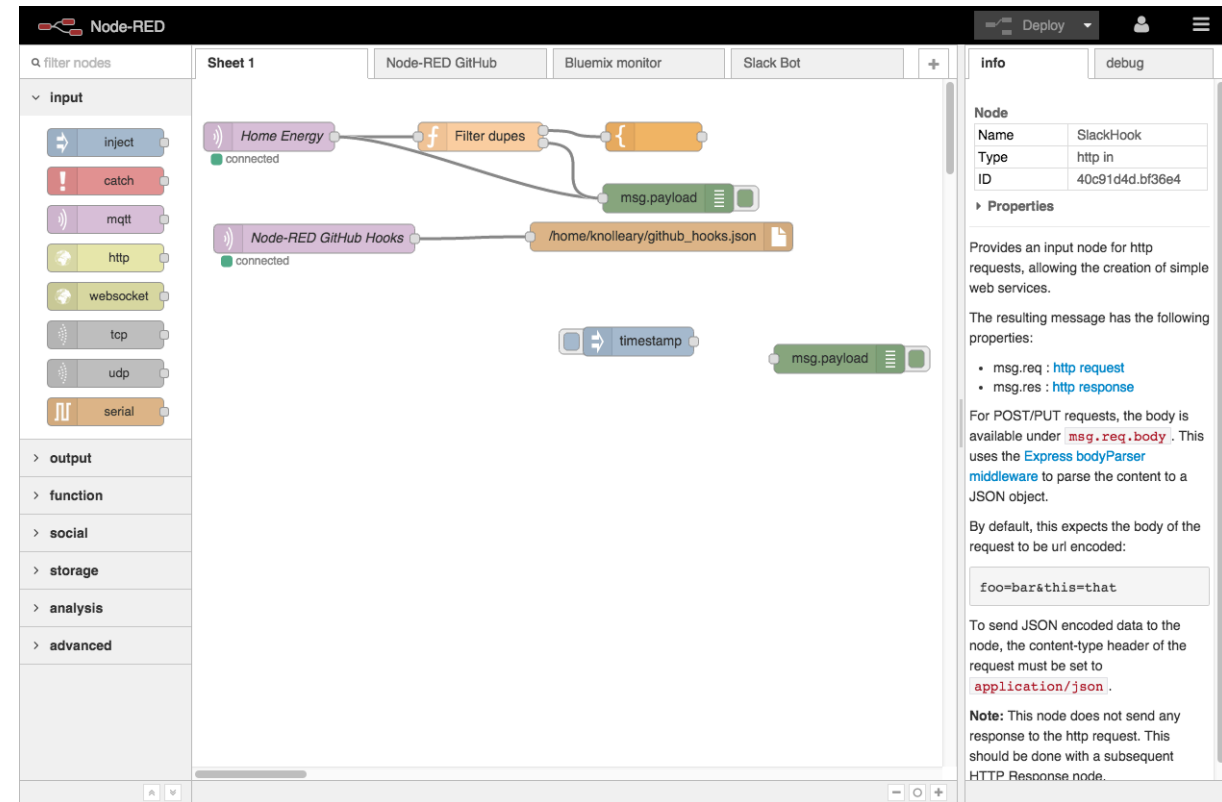
- Core contributor of Node-RED project
- Organizer of Node-RED User Group
- Member of LF AI & Data Foundation Community

<https://openjsf.org/blog/node-red-unleashed>

What is Node-RED?

Low-code development tool from novice users to IT professionals

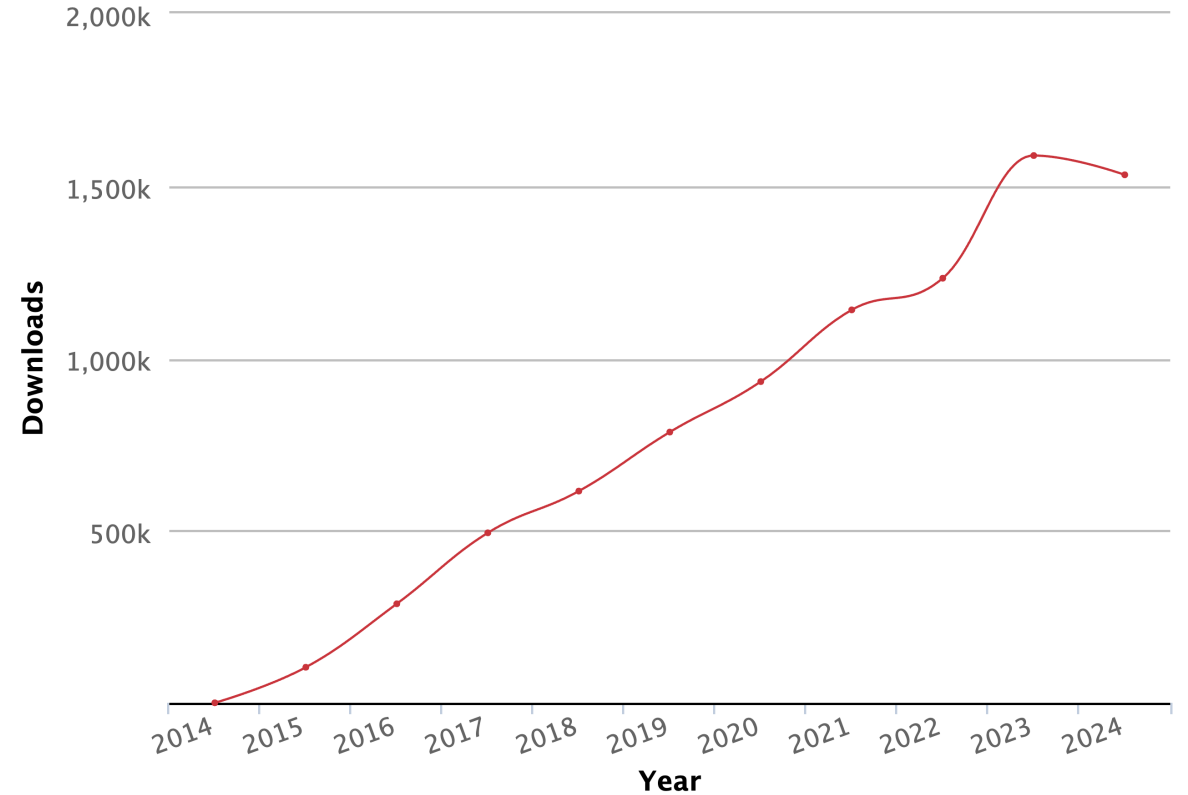
- Released by IBM as an open source software
- Hosted by the OpenJS Foundation
- Used in manufacturing industry to realize industrial IoT



Node-RED statistics

Node-RED has been widely used for 10 years.

- 155 releases
- 9,802 commits
- 237 contributors
- 22k GitHub stars
- 3B+ Docker pulls
- 5,562 community nodes



Supported Environments

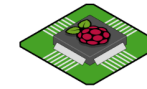
Node-RED supports environments that can run Node.js.

- Local PC
 - Windows
 - macOS
 - Ubuntu
- Docker
- Raspberry Pi
- Cloud



Running locally

Installing Node-RED on your local computer



Raspberry Pi

Get started using our all-in-one install script for the mighty Raspberry Pi



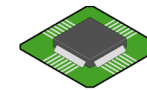
Docker

Running Node-RED using Docker



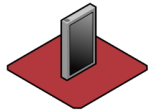
Install from git

Building Node-RED from source. Get the very latest development code and start contributing.



BeagleBone Boards

Running Node-RED on BeagleBone boards



Android

A bit experimental, but you can run on Android devices using Termux



FlowFuse

Running a multi-tenant Node-RED solution



AWS

Get started running on Elastic Beanstalk or EC2



Microsoft Azure

Running on an Azure Virtual Machine instance

How to install

- Local environment

```
sudo npm install -g --unsafe-perm node-red  
node-red
```

- Docker

```
docker run -it -p 1880:1880 --name  
mynodered nodered/node-red
```

- Raspberry Pi

```
bash <(curl -sL  
https://raw.githubusercontent.com/node-red/linuxinstallers/master/deb/update-nodejs-and-nodered)
```

Node-RED nodes

Many predefined and custom nodes for services and devices

- Data collection
- Machine control
- External server access
- Database connection
- Data analysis
- Visualization

The screenshot displays the Node-RED library interface. At the top, there's a navigation bar with links: home, about, blog, documentation, forum, flows, and github. Below this is a search bar labeled 'Search library'. On the left side, there's a 'Filter' section with checkboxes for 'Nodes' (checked), 'Flows' (checked), and 'Collections' (checked). Below the filter is a 'Category' section with checkboxes for various categories: AI, Communication, Data & Analytics, Database, Date & Time, Hardware, Home Automation, Industrial, Security, Storage, Tools, UI, and Utility. The main area shows a grid of node cards, each with a title, description, version, download count, star rating, and a 'node' button. The nodes are sorted by 'Downloads'.

Node Name	Description	Version	Downloads	Stars	Label
node-red-dashboard	A set of dashboard nodes for Node-RED	v3.6.6	19282	4.6	node
node-red-node-ui-table	Table UI widget node for Node-RED Dashboard	v0.4.5	13481	4.7	node
node-red-contrib-knx-ultimate	Control your KNX and KNX Secure installation via Node-Red! A bunch of KNX nodes, with integrated Philips HUE control and ETS group address importer. Easy to use and highly configurable.	v4.0.10	12168	4.9	node
@flowfuse/node-red-dashboard	FlowFuse Dashboard - A collection of Node-RED nodes that provide functionality to build your own UI applications (inc. forms, buttons, charts).	v1.28.0	9160	4.8	node
node-red-contrib-credentials	Provides a credentials node to store one or more private values; preventing export to flows or version control.	v0.2.3	8506	4.5	node
node-red-contrib-calc	A Node-Red node to perform basic mathematical calculations	v1.0.6	8235	4.4	node
node-red-iot-mqtt-api	return right result to node-red	v1.0.1	7875	5.0	node
test-switchbot-devices	A sample node for switchbot	v0.0.14	7813		node
node-red-contrib-dashboard-average-bars	Calculate and display the average values of msg.payload in a bar chart.	v0.0.6	7805	3.0	node
node-red-contrib-moment	Node-Red Node that produces formatted Date/Time output using the MomentJS library. Timezone, dst and locale aware.	v5.0.0	7348	4.3	node
node-red-contrib-tableify	Uses npm package tableify to convert json into an html table. Returns the html in msg.payload	v0.0.2	6538	3.7	node
node-red-contrib-string	Provides a string manipulation node with a chainable UI based on the concise and lightweight stringjs.com.	v1.0.0	5850	4.6	node
node-red-node-email	Node-RED nodes to send and receive simple emails.				
node-red-contrib-modbus	The all in one Modbus TCP, UDP and Serial contribution long term supported package for Node-RED.				
node-red-contrib-image-tools	A dynamic image editor, image viewer, 1D/2D Barcode Generator and 1D/2D Barcode Decoder nodes				

Pre-installed Devices

Many devices equip Node-RED as a pre-installed software.

- Seeed Studio, reTerminal DM, reCamera
- Advantech, ADAM-6700
- ADLINK, Vizi-AI Devkit
- QNAP, QIoT Suite
- Emerson, PACEdge

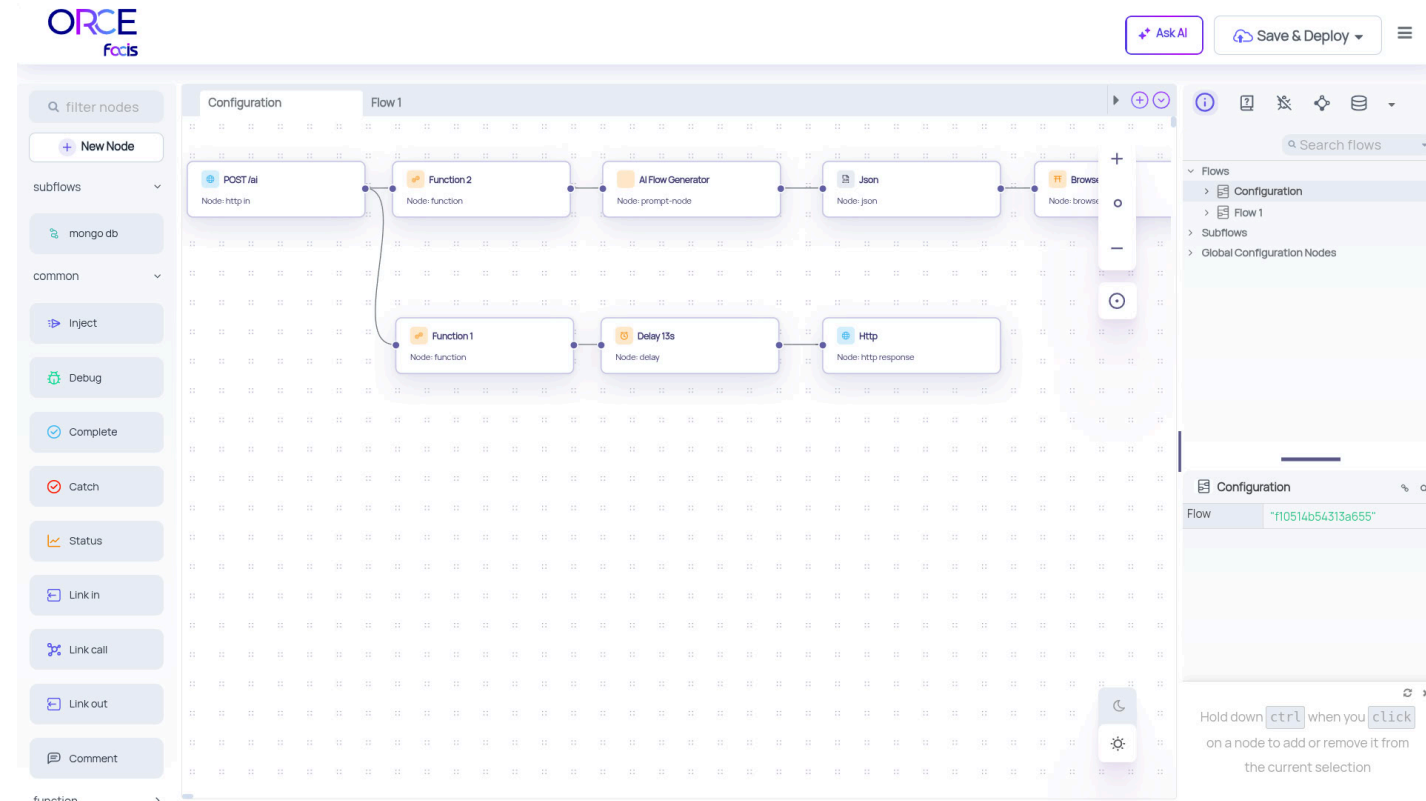


GAIA-X Federation Services

OSS set for data exchange between companies

- Hosted by the Eclipse Foundation from 2023
- GXFS Workflow Engine based on Node-RED

<https://github.com/eclipse-xfsc/orchestration-engine>

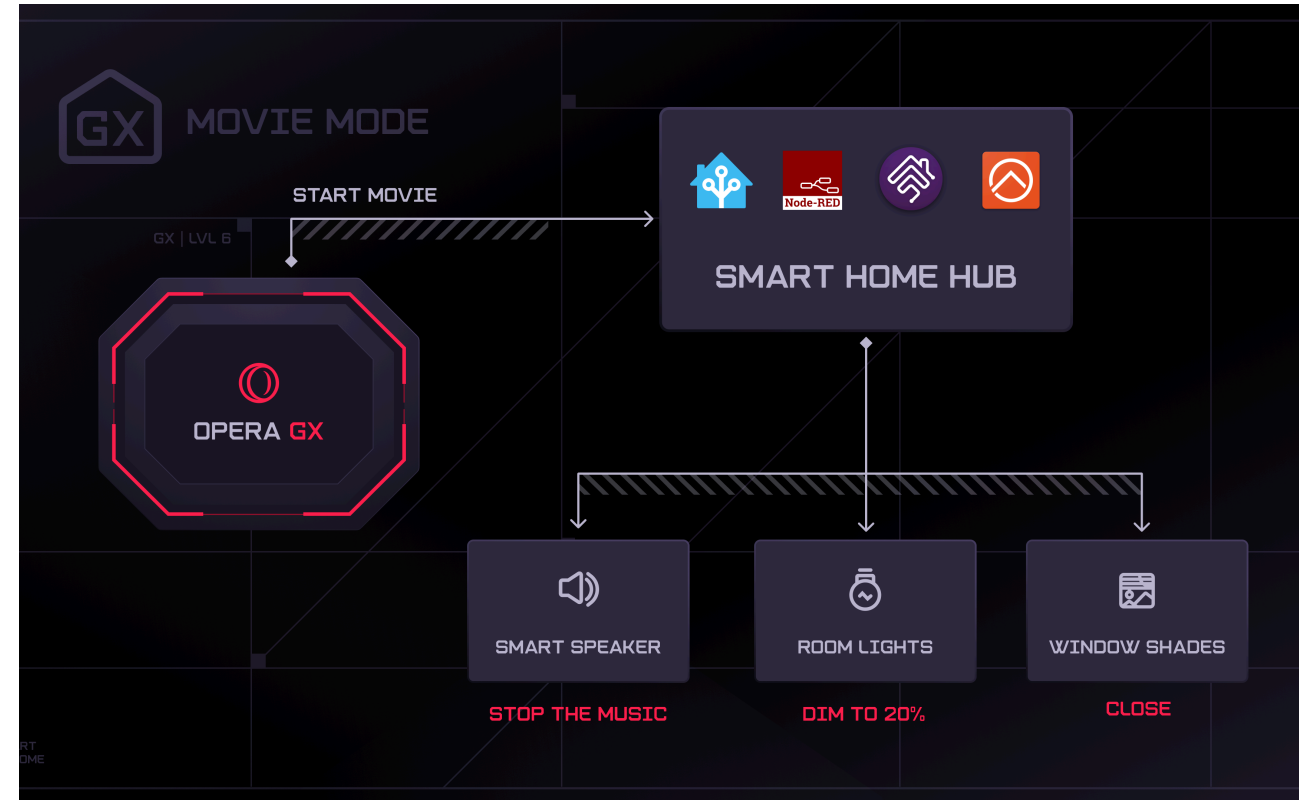


Opera Browser

Opera Software adopted Node-RED for their browser, Opera GX for smart home use cases

- Getting your room ready for watching movies
- Syncing the colors of smart bulbs with the theme of Opera GX

<https://github.com/operasoftware/opera-smart-home>



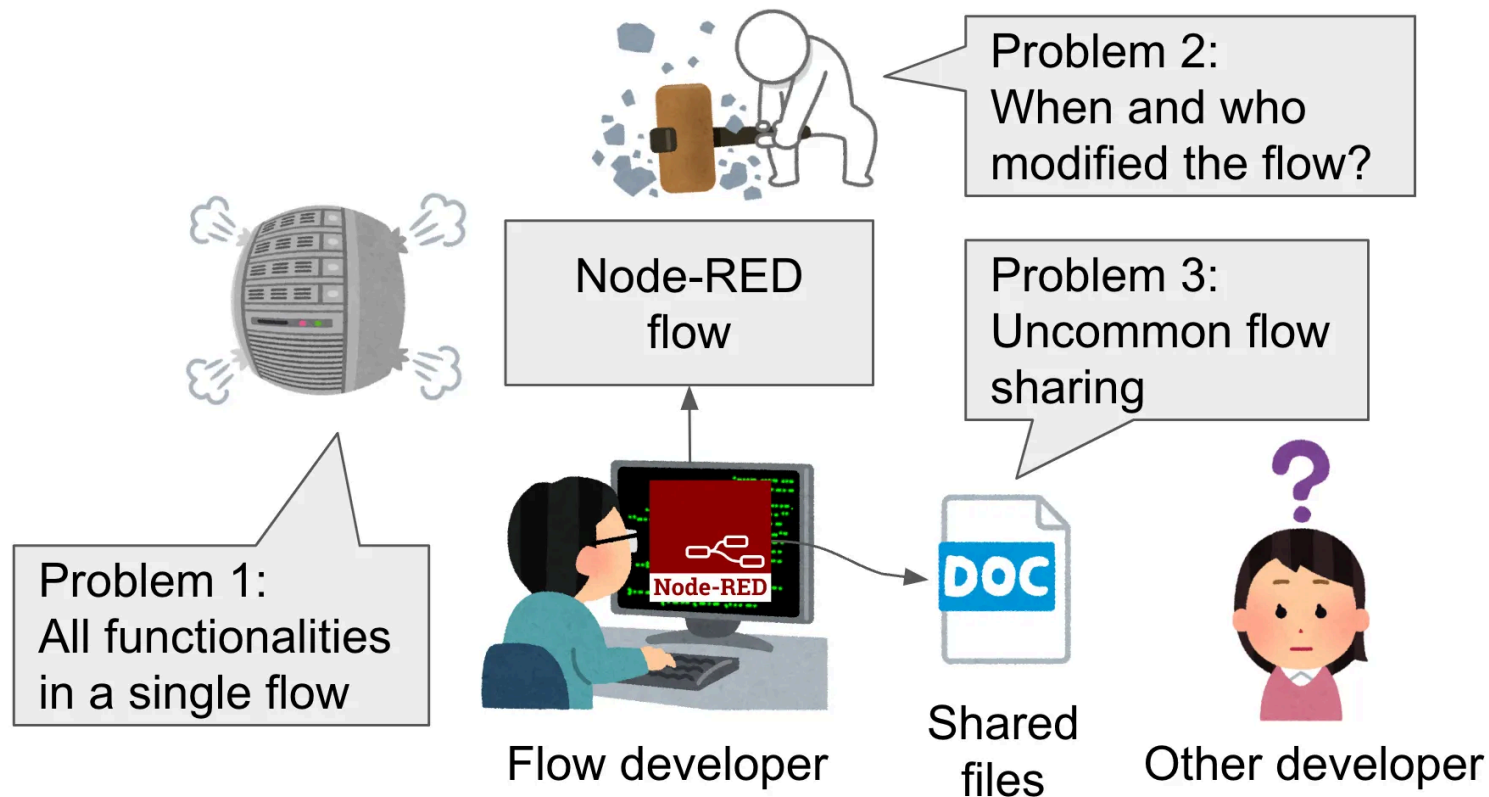
Advanced features

Features to improve developer experiences

- Project feature (Git functionality)
 - Managing flows
 - Tracking changes
- Flow Debugger
- Flow Linter

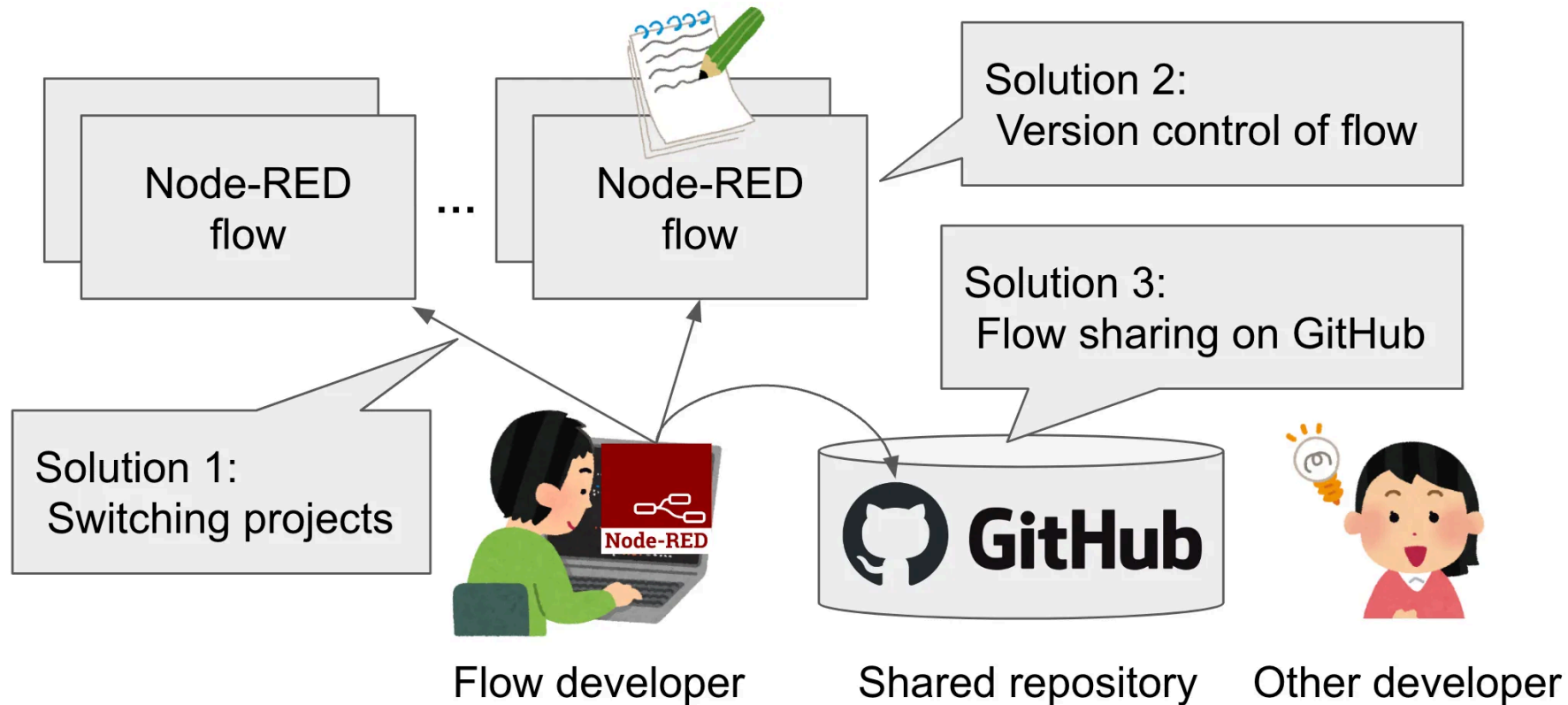
Issues without project feature

We encounter the following issues in default setting.



Solution with project feature

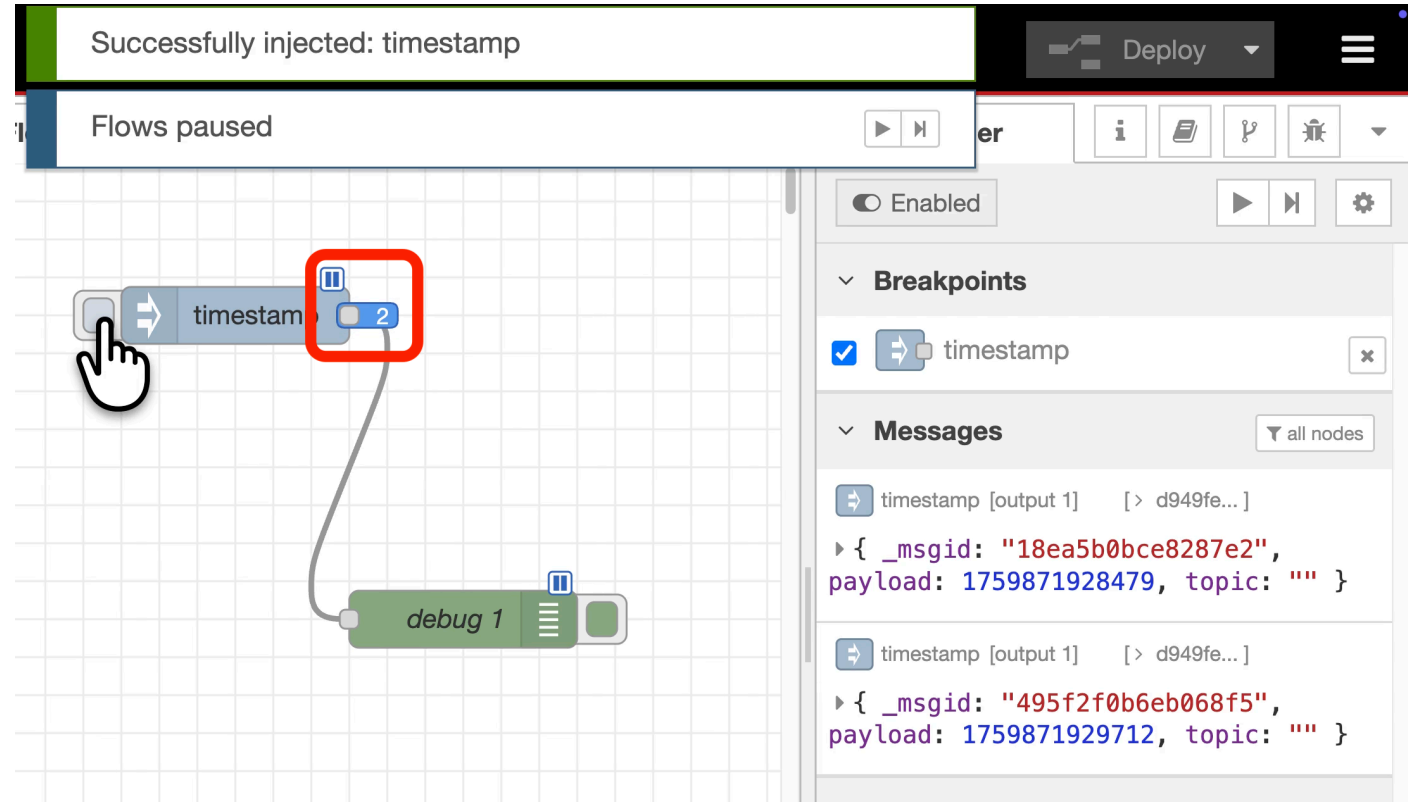
Project feature solves the previous issues.



Flow Debugger

Provides same user experience as code debugger

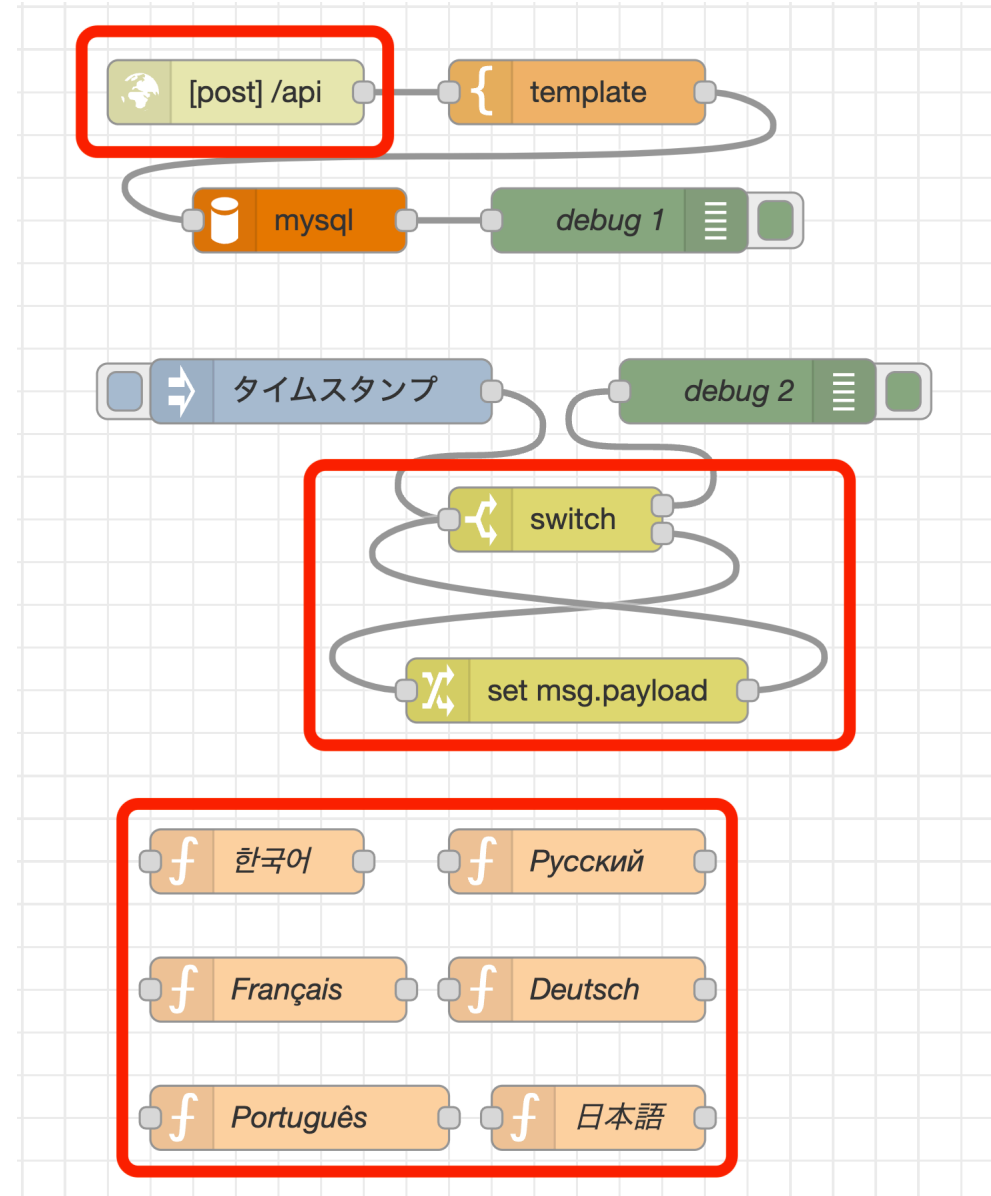
- Creating breakpoints
- Pausing flows
- Inspecting messages
- Step by step execution



Flow Linter

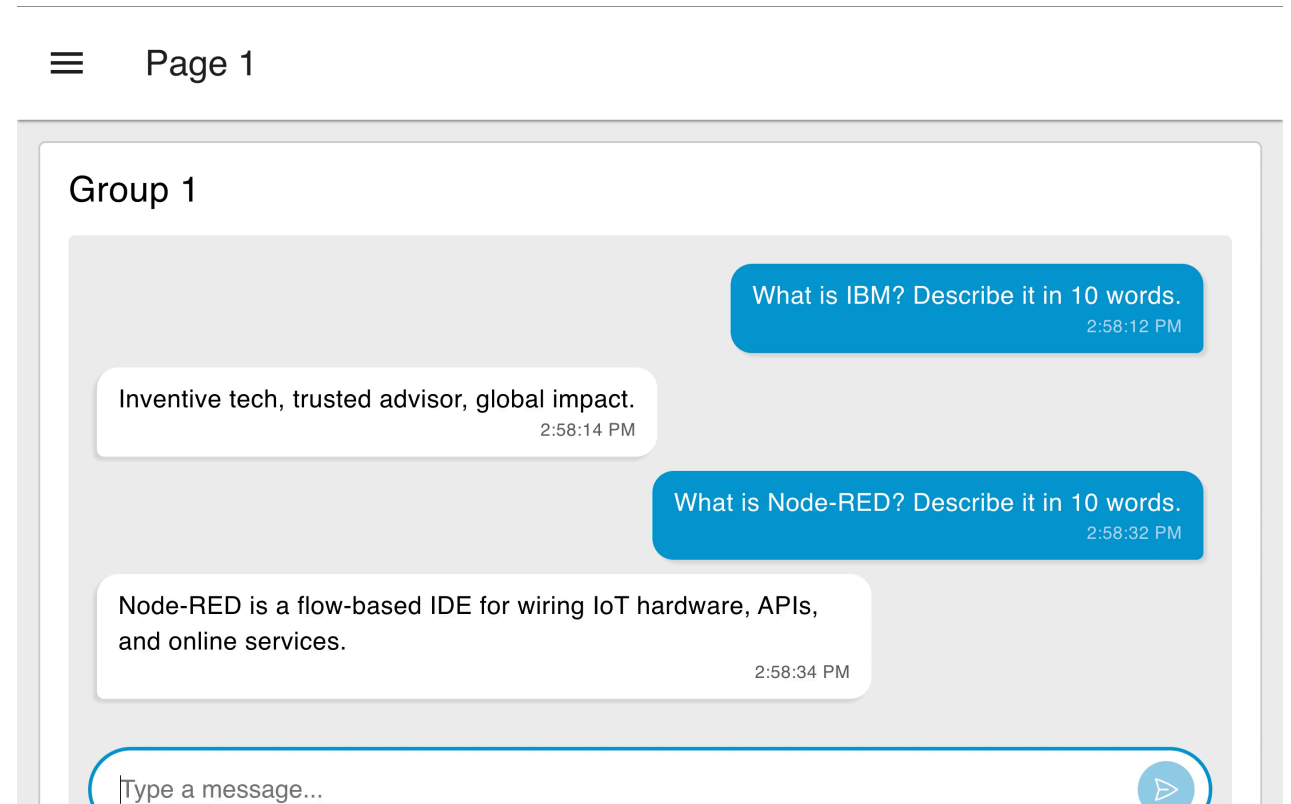
Provides real-time feedback same as code linter

- Supported potential issues:
 - http-in node without http-response node
 - Looping flows
 - Overlapping nodes
 - Unset node names
- Custom rules:
 - Recommendation of English node names



Chat application with Granite model

- Using IBM's Granite model on Ollama
- Chat UI with Node-RED Dashboard



Conclusion

- I introduced the advanced features of Node-RED
 - Project feature
 - Flow Debugger
 - Flow Linter
- Development example of Chat UI application

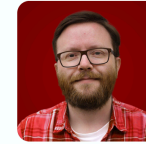
-> I believe that you will develop the great flows using these features and example.

Node-RED Conference 2025

- Date: Tuesday, November 4, 2025
- Location: Online
- Speakers: Intel, FlowFuse, Victron Energy, Fluidy Financial Services, City of Morro Bay / GCIS and more
- Registration: Free

<https://nrcon.nodered.org/>

Speakers



Nick O'Leary 🌱
Node-RED Co-creator / FlowFuse
CTO



ZJ van de Weg 🌱
CEO
FlowFuse



Felix Reck
IT Application Manager



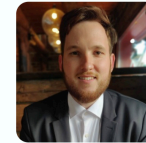
Zepu Li



Christian Lagerkvist
Solutions Architect
Fluidy Financial Services



Dirk-Jan Faber
Software Engineer
Victron Energy



Øivind Heggland
Technical Project Manager



Alexander Kuesel
Operational Technology Software
Development Manager
Intel



Anton Rodriguez-Dmitriev
Operational Technology (OT)
Controls Engineer
Intel



Nicolas Alvarez Garcia
Industrial Control Systems
Engineer
Intel



Robert Clynych
Operational Technology (OT)
engineer
Intel



Josh Dudley
Manufacturing and Automation
Engineer



Grant Chase
Operational Technology Specialist
City of Morro Bay / GCIS